

A Functional-Logical Derivation of the Hubble Tension Regime-Dependent Relational Expansion in a Finite-Stabilization Ontology

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1 Purpose

This document derives, in a functional-logical manner, why two different observational inference pipelines can yield different effective Hubble parameters H_0 without introducing a new fundamental force or an additional ontological primitive.

The central hypothesis is:

Cosmic “expansion” is not the stretching of a geometric substrate, but a regime-dependent change in relational capacity among energy-density distributions under finite stabilization propagation.

Under this view, the “Hubble tension” is not a geometric inconsistency but a *regime-aggregation mismatch*: different measurements probe different relational regimes and apply different coarse-graining operators.

2 Layer I: Ontological primitives (fixed)

We admit only:

- Energy E as transition capacity,
- Mass as bound energy density,
- Relation as physical relational structure (“space” as relation, not container),
- Process Φ as stabilization operator with finite propagation limit c_Φ (“time” as process, not dimension).

No extra “expansion force” is postulated.

3 Layer II: Minimal cosmological projection (commitments)

3.1 C1: Expansion as relational capacity change

Define a relational scale descriptor $a(t)$ as a *projection variable* that summarizes average relational separation capacity among energy-density distributions. This is not a geometric fabric; it is an effective descriptor of relational organization.

Define:

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)}.$$

3.2 C2: Regime dependence

Let $\rho(t, \mathbf{x})$ denote (effective) energy density and let the framework admit that the functional relationship between expansion and density is *nonlinear* because stabilization and binding do not scale linearly across regimes.

We therefore define a minimal regime map:

$$H_{\text{eff}}(t, \mathbf{x}) = \mathcal{H}(\rho(t, \mathbf{x}); \rho_c, c_\Phi),$$

where ρ_c is a fixed critical density scale separating dominance regimes, and c_Φ bounds propagation of stabilization.

Interpretation. For $\rho \gg \rho_c$, binding/stabilization dominates and relational capacity growth is suppressed. For $\rho \ll \rho_c$, transition-dominant freedom in relations increases and the effective relational capacity growth can be larger.

3.3 C3: Minimal monotone form (no new force)

To keep the projection minimal and falsifiable, assume only:

1. $\mathcal{H}(\rho)$ is positive and continuous in the relevant domain,
2. $\mathcal{H}$ changes curvature across ρ_c (nonlinear regime transition),
3. no event-by-event tuning is permitted once ρ_c and the functional family are fixed.

A minimal example family (illustrative, not yet fitted) is:

$$\mathcal{H}(\rho) = H_\star f\left(\frac{\rho}{\rho_c}\right), \quad f(u) = (1 + u^\alpha)^{-\beta}, \quad \alpha, \beta > 0,$$

where $H_\star$ is a global normalization and α, β are fixed once in the theory package. (Other fixed families are possible; what matters for the logic is *nonlinearity* and a *regime scale* ρ_c .)

4 Why different inference pipelines yield different effective H_0

4.1 Key lemma: nonlinear averaging mismatch

Let X be a random variable representing the density encountered by a given measurement pipeline (e.g. along a line of sight or across a sky patch). If $\mathcal{H}$ is nonlinear, then in general:

$$\mathcal{H}(\mathbb{E}[X]) \neq \mathbb{E}[\mathcal{H}(X)].$$

This is the sole mathematical mechanism required to generate two effective “ H_0 ” values from the same underlying relational ontology.

4.2 Pipeline A: Early-universe (CMB) as global-regime inference

The CMB-based inference effectively constrains an early-time global state by fitting a model that assumes a single global expansion function and extrapolates to $z \approx 0$.

Functionally, its inference resembles:

$$H_0^{(\text{CMB})} \approx \mathcal{A}_{\text{CMB}}[\mathcal{H}(\rho)],$$

where $\mathcal{A}_{\text{CMB}}$ is a global aggregation operator dominated by near-homogeneous, high-redshift conditions and model-dependent integrals.

If the assumed aggregation implicitly approximates:

$$\mathcal{A}_{\text{CMB}} \approx \mathcal{H}(\langle \rho \rangle_{\text{early}}),$$

then it corresponds to evaluating $\mathcal{H}$ near a regime associated with early-universe densities.

4.3 Pipeline B: Late-universe distance ladder as local-regime inference

The late-universe distance ladder probes present-day, *strongly inhomogeneous* structure: voids, filaments, halos, clusters. Its inference resembles a different operator:

$$H_0^{(\text{SN})} \approx \mathcal{A}_{\text{SN}}[\mathcal{H}(\rho)],$$

where $\mathcal{A}_{\text{SN}}$ is a line-of-sight and sample-selection weighted aggregation over late-time inhomogeneous density fields:

$$\mathcal{A}_{\text{SN}} \approx \langle \mathcal{H}(\rho(t_0, \mathbf{x})) \rangle_{\text{SN weights}}.$$

4.4 Functional conclusion: two effective H_0 values are expected

If (i) $\mathcal{H}$ is nonlinear with a regime transition near ρ_c , and (ii) $\mathcal{A}_{\text{CMB}}$ and $\mathcal{A}_{\text{SN}}$ weight different density regimes, then generically:

$$H_0^{(\text{CMB})} \neq H_0^{(\text{SN})}.$$

This is not a contradiction of “geometry” in this ontology, because geometry is not primary. It is a difference in *what is being averaged* and *how* it is aggregated.

5 Why no additional “repulsive force” is required

In this framework, the apparent “repulsion” associated with accelerated expansion is not a new force. It is a relational-regime outcome:

- Gravity corresponds to binding dominance at higher effective density and stronger coupling.
- Expansion corresponds to transition-dominant relational capacity growth when binding dominance weakens under lower effective density and finite propagation constraints.

The system does not require a counter-force; it requires only:

- nonlinear regime dependence $\mathcal{H}(\rho)$,
- inhomogeneous density fields at late times,
- distinct aggregation operators for different observation pipelines.

6 Testable predictions (pre-registered form)

Once a specific functional family for $\mathcal{H}$ is fixed (e.g. by independent constraints), the following predictions become testable without adding new primitives:

6.1 P1: Environment dependence

Local effective expansion inferred from distance-ladder samples should correlate with environmental density statistics (void-weighting vs cluster-weighting).

6.2 P2: Consistency under matched aggregation

If one reprocesses both pipelines with a *common* aggregation operator that explicitly accounts for regime dependence, the inferred H_0 discrepancy should reduce.

6.3 P3: Redshift-dependent transition signature

A regime transition near ρ_c implies a characteristic change in inferred $H(z)$ slopes across the redshift interval where the effective density field crosses the transition band.

7 Explicit boundaries of claim

This document does *not* claim a fully specified numerical value for the tension. To produce numerical predictions one must freeze:

- the functional family $f(\rho/\rho_c)$,
- the critical scale ρ_c ,

- and the precise aggregation operators $\mathcal{A}_{\text{CMB}}$ and $\mathcal{A}_{\text{SN}}$ (including selection weights).

The functional-logical result established here is structural:

Regime-dependent nonlinear expansion+different aggregation operators $\Rightarrow$ different effective H_0 estimates.

8 Conclusion

If space is relational and time is stabilization process Φ with finite propagation, then “expansion” is a regime-dependent relational capacity change rather than a geometric stretching. Under nonlinear regime dependence and distinct inference operators, a Hubble tension is not only possible, but generically expected, without postulating new forces or new ontological primitives.